Firewalls and IDS/IPS

IE3122: Network Security | Week 7

What a firewall actually is, how it evolved, and the seven types you'll encounter

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Today's Agenda

- What a firewall is, and how it compares to IDS/IPS
- Software versus hardware firewalls
- A brief history of firewalls
- Stateless versus stateful inspection
- Seven firewall types, in the order they evolved, each with mechanism, trade-offs, and use cases
- The evolution of firewalls, revisited analytically
- A full comparison summary

Learning Outcomes for Today

LO3

– Analyse security requirements and recommend appropriate technologies.

LO4

– Evaluate different security technologies, compare them, and identify appropriate use cases.

– *By the end of this session you should be able to name, distinguish, and recommend a firewall type for a given scenario, and explain why it evolved the way it did.*

Foundational Concepts

What a firewall is, and what it deliberately is not

What Is a Firewall?

- A firewall is a network security device or piece of software that monitors and controls incoming and outgoing traffic based on a predetermined set of rules, acting as a barrier between a trusted network and an untrusted one.
- Its core function is fundamentally binary per rule: allow or deny. Whether that decision is based on a simple port number or deep content inspection, every firewall type this week ultimately reduces to that same choice.
- A firewall enforces a policy. It doesn't learn or adapt on its own. It does exactly what its rule set tells it to do, which is precisely why the rule set itself matters so much.

What Is IDS/IPS? (A Brief Introduction)

- An Intrusion Detection System (IDS) monitors network or system activity for suspicious behaviour or known attack signatures, and generates an alert when it finds something. Critically, an IDS does not block traffic itself.
- An Intrusion Prevention System (IPS) does everything an IDS does, but sits inline and can actively block or drop malicious traffic in real time, much like a firewall does, except its decision is based on behavioural or signature analysis rather than a simple allow/deny rule.
- This is intentionally a brief introduction. IDS/IPS gets its own full treatment, including detection methods and placement, in a later week. For now, you only need enough to clearly distinguish it from a firewall.

Firewalls vs. IDS/IPS: Similarities and Differences

- Similarities: both are network security controls, both can be deployed inline, both rely on a rule or signature set to make decisions, and both are components of a layered defence rather than a complete solution on their own.
- A firewall is primarily preventive, making decisions from explicit allow/deny rules (source and destination address, port, protocol, and for advanced types, application or content). It generally isn't looking for attack patterns, just policy compliance.
- An IDS is purely detective. It analyses traffic for attack signatures or anomalies and generates alerts, with no ability to block anything itself.
- An IPS combines an IDS's detection method with a firewall's blocking action: inline detection plus active response. A simple way to remember it: a firewall is a locked door checking IDs against a guest list, an IDS is a camera that notices suspicious behaviour and alerts a guard, and an IPS is a guard who can also step in.

Software vs. Hardware Firewalls

- A hardware firewall is a dedicated physical appliance deployed at the network perimeter, protecting an entire network segment at once. It typically offers high throughput (often via dedicated processing hardware), doesn't consume any host's resources, and centralises policy in one place. The trade-off is higher upfront cost, and it generally can't see or control traffic happening entirely within the network it protects.
- A software firewall runs as an application or service on an individual host, such as a built-in OS firewall or a virtualised instance. It offers granular, per-host control and is cheap or free to deploy widely, but it consumes that host's own CPU and memory, and can be disabled or bypassed if the host itself is compromised.
- In practice, real networks use both together: a hardware firewall at the perimeter plus software firewalls on individual endpoints. This is the same layered-defence principle that has run through every week of this module.

A Brief History of Firewalls

- Late 1980s: the first packet filtering firewalls appear, examining only IP addresses, ports, and protocol in each packet's header, with no awareness of connection state at all.
- 1994: Check Point introduces stateful inspection with FireWall-1, a major leap that tracks the state of active connections rather than judging every packet in total isolation.
- Through the mid-to-late 1990s, circuit-level gateways and application-layer proxy firewalls develop in parallel, each trying to see further into a connection than stateful inspection alone could.
- Through the 2000s, Unified Threat Management (UTM) appliances emerge, bundling firewall, antivirus, IDS/IPS, and content filtering into a single box.
- From roughly 2007 to 2010 onward, Next-Generation Firewalls (NGFW) emerge, credited significantly to Palo Alto Networks, offering deep packet inspection, application awareness, and integrated intrusion prevention in one tightly-built system, alongside the parallel rise of Web Application Firewalls (WAF) as a web-specific specialisation.
- This is purely the timeline for now. Once you've seen all seven types in detail, we'll come back and analyse why each generation actually emerged.

CORE CONCEPT

Stateless vs. Stateful Inspection

- Stateless inspection evaluates every packet entirely on its own, checking only that packet's own header fields against a static rule list. It has no memory of any packet that came before it.
- This means a stateless firewall cannot tell the difference between a legitimate reply to a connection your own network initiated, and an unsolicited or spoofed packet crafted to look the same. Recall Week 6: this is exactly the kind of blind spot several TCP attacks exploit.
- Stateful inspection maintains a state table tracking every active connection: source and destination address and port, protocol, and connection stage. Once a connection is legitimately established, return traffic for it is automatically permitted, without needing a separate explicit rule for the reply direction.
- This single distinction, whether a firewall remembers anything about the conversation it's part of, is the deciding factor behind several of the types you're about to see. Keep it in mind as we go through them.

Seven Firewall Types

In the order they historically evolved

Seven Types at a Glance

Type	OSI Layer	State Awareness
Packet Filtering	Network / Transport (L3/L4)	Stateless
Stateful Inspection	Network / Transport (L3/L4)	Stateful
Circuit-Level Gateway	Session (L5)	Stateful (session only)
Application-Layer / Proxy	Application (L7)	Stateful, protocol-aware
Next-Generation Firewall (NGFW)	L3 through L7, integrated	Stateful, application-aware, deep inspection
Web Application Firewall (WAF)	Application (L7), web-specific	Stateful, web-attack-pattern-aware
Unified Threat Management (UTM)	L3 through L7, bundled modules	Stateful, multiple bundled functions

TYPE 1

Packet Filtering Firewall: Mechanism

- Operates at the Network and Transport layers, examining each packet's header entirely in isolation: source and destination IP address, source and destination port, and protocol.
- Compares each packet against a static Access Control List processed top to bottom, stopping at the first matching rule.
- Stateless: it has no awareness of whether an incoming packet is genuinely part of an existing, legitimate conversation.
- Because it never inspects payload content, only headers, it is the fastest and lowest-overhead type covered today.

TYPE 1

Packet Filtering Firewall: Advantages and Disadvantages

Advantages

- Very fast, with minimal processing overhead
- Simple to understand and configure for basic policies
- Inexpensive
- Effective as a first line of defence directly on a router

Disadvantages

- No awareness of connection state
- Cannot inspect payload content, so application-layer attacks pass through
- Rule sets can grow complex and hard to maintain at scale
- Fragmented packets can evade filtering

TYPE 1

Packet Filtering Firewall: Use Cases

- Router-based Access Control Lists, exactly what you'll configure hands-on in this week's lab.
- First-line perimeter filtering for straightforward protocol and port-based blocking.
- High-throughput environments where raw speed matters more than deep inspection.
- Small networks and home routers, where simplicity is the priority.

TYPE 2

Stateful Inspection Firewall: Mechanism

- Operates at the same layers as packet filtering, but maintains a state table tracking every active connection: addresses, ports, protocol, and TCP sequence and flag information.
- New connection attempts are evaluated against the rule set, but return traffic belonging to an already-established, legitimate connection is automatically permitted without a separate rule.
- Can detect out-of-state anomalies, for example a packet claiming to belong to a TCP session that was never properly established, directly relevant to Week 6's hijacking and RST injection discussion.

TYPE 2

Stateful Inspection Firewall: Advantages and Disadvantages

Advantages

- Substantially stronger security than pure packet filtering, since decisions have context
- Simpler rule sets, since only the initiating direction needs an explicit rule
- Better resistance to several spoofing and scanning techniques

Disadvantages

- More resource-intensive, since the state table must be maintained in memory
- The state table itself can be exhausted by a flood of half-open connections (SYN flood)
- Still generally cannot inspect encrypted payload content
- More complex to troubleshoot than simple packet filtering

TYPE 2

Stateful Inspection Firewall: Use Cases

- The default assumption behind most modern commercial firewall appliances.
- Standard enterprise perimeter firewalls, where solid protection is needed without the overhead of full application-layer inspection.
- Any environment needing genuine context-aware filtering without paying for NGFW-level capability.

Circuit-Level Gateway: Mechanism

- Operates at the Session layer, verifying that a TCP handshake and session setup is legitimate, for example monitoring the SYN, SYN-ACK, and ACK sequence, before allowing the session to proceed.
- Once a session is validated, it does not inspect the actual payload content at all. It checks that the session was formed correctly, not what flows through it afterward.
- SOCKS proxy is a classic real-world implementation of this approach.

Circuit-Level Gateway: Advantages and Disadvantages

Advantages

- Lower overhead than full application-layer inspection
- Hides internal network addressing from the outside
- Relatively fast once a session is validated

Disadvantages

- No content-level filtering at all after the handshake
- Malicious payload inside an already-validated session passes through undetected
- Limited standalone use today, largely absorbed into later firewall generations

Circuit-Level Gateway: Use Cases

- Historically used in early combined proxy-firewall products.
- SOCKS-based application proxying specifically.
- Scenarios needing to hide internal network topology from external parties while keeping overhead low.
- Rarely deployed as a sole, standalone firewall today. More of a conceptual building block whose ideas live on inside later firewall generations.

TYPE 4

Application-Layer (Proxy) Firewall: Mechanism

- Operates at the Application layer, acting as a genuine intermediary: the client never connects directly to the destination. The firewall terminates the client's connection, inspects the request in full application-protocol detail, then opens a separate connection to the real destination on the client's behalf.
- Because it fully understands the specific application protocol involved, for example HTTP methods or FTP commands, it can enforce very granular rules and even strip malicious content from a response before forwarding it.

TYPE 4

Application-Layer (Proxy) Firewall: Advantages and Disadvantages

Advantages

- Deepest inspection and control among the classic types
- Strong anonymisation, client's real address never exposed
- Can actively sanitise malicious content, not just block it

Disadvantages

- Highest performance overhead of the classic types
- Needs a dedicated proxy module for every application protocol
- Can become a bottleneck or single point of failure under heavy load

Application-Layer (Proxy) Firewall: Use Cases

- Environments requiring very strict content control, such as corporate web or email proxies filtering specific content types.
- Situations needing strong anonymisation of internal client identities.
- Legacy enterprise proxy deployments, such as Squid-based web proxies.
- The direct conceptual foundation behind modern WAFs, which apply the same deep-inspection principle specifically to web traffic.

Next-Generation Firewall (NGFW): Mechanism

- Integrates traditional stateful inspection with deep packet inspection, application awareness (identifying traffic by the actual application in use, not just by port number), integrated intrusion prevention, and often user-identity-based policy tied to specific users or groups rather than just IP addresses.
- Built as a single-pass, tightly integrated architecture from the ground up, in deliberate contrast to UTM's bundled-separate-modules approach, covered shortly.

TYPE 5

Next-Generation Firewall (NGFW): Advantages and Disadvantages

Advantages

- Comprehensive, modern protection combining many functions in one system
- Application-level visibility gives far more precise policy than port or protocol alone
- Often able to decrypt and inspect SSL/TLS traffic
- Strong fit against sophisticated, multi-vector modern attacks

Disadvantages

- Significantly more expensive than earlier generations
- Higher configuration and management complexity
- Deep inspection, especially SSL decryption, can impact throughput
- Carries a large amount of security responsibility in one place

TYPE 5

Next-Generation Firewall (NGFW): Use Cases

- Modern enterprise perimeter security, effectively the current default standard.
- Organisations needing application-aware policy, such as allowing general web browsing while blocking specific risky applications regardless of the port they use.
- Environments with significant encrypted traffic needing genuine SSL inspection capability.
- Any organisation wanting one consolidated, tightly integrated security platform rather than several separate products.

Web Application Firewall (WAF): Mechanism

- Operates at the Application layer, but specialised specifically for HTTP and HTTPS web traffic rather than general-purpose like an NGFW.
- Sits in front of web applications and servers, inspecting requests and responses specifically for web-application attack patterns: SQL injection, cross-site scripting, cross-site request forgery, and other OWASP Top 10-style threats.
- Can use signature-based detection (matching known attack patterns), a positive security model (allowing only known-good request patterns and denying everything else), or both together.

TYPE 6

Web Application Firewall (WAF): Advantages and Disadvantages

Advantages

- Highly specialised, effective against web-application-specific attacks an NGFW typically wouldn't catch
- Can be deployed close to the application, including as a reverse proxy or cloud-based service

Disadvantages

- Narrow scope, protects web applications only
- Needs to be paired with a broader firewall for general network protection
- Rule tuning can be difficult, with real risk of false positives or false negatives

TYPE 6

Web Application Firewall (WAF): Use Cases

- Any organisation running public-facing web applications, such as e-commerce platforms or customer portals, exactly the kind of system referenced throughout this module's Layer 3 and Layer 4 scenarios.
- Compliance-driven environments. PCI-DSS, for example, commonly expects WAF protection for payment-processing web applications.
- Cloud-hosted applications, where a cloud-based WAF service is a common and easily deployed addition.

Unified Threat Management (UTM): Mechanism

- A single hardware or virtual appliance that bundles multiple distinct security functions together: firewall, antivirus, IDS/IPS, content and URL filtering, VPN, and often spam filtering.
- The key distinction from NGFW: UTM typically runs these as separate, largely independent modules within one box, a management and deployment convenience, whereas NGFW is built from the ground up around one tightly integrated inspection engine.

TYPE 7

Unified Threat Management (UTM): Advantages and Disadvantages

Advantages

- Simplifies management, a single appliance, vendor, and interface
- More cost-effective than buying and integrating multiple standalone products
- Easier initial deployment, particularly for smaller organisations

Disadvantages

- Can create a performance bottleneck under heavy load
- Single point of failure for many security functions at once
- May not match the depth of best-of-breed standalone products
- Scaling can be harder than with more modular architectures

Unified Threat Management (UTM): Use Cases

- Small and medium-sized businesses wanting comprehensive security without managing many separate vendors and products.
- Branch office deployments, where simplicity of management matters more than maximum possible performance.
- Organisations with limited dedicated security staff, who benefit from one consolidated management interface rather than several.

Bringing It Together

Why each generation emerged, and how it all compares

ANALYSIS

The Evolution of Firewalls, Revisited

- Packet filtering came first simply because early networks needed basic access control and processing power was limited. Simple header matching was what was feasible.
- Stateful inspection emerged specifically because packet filtering's lack of context created real, exploitable blind spots, such as being unable to distinguish legitimate return traffic from spoofed traffic. This was a genuine security improvement, not just a performance one.
- Circuit-level gateways and application-layer proxies developed in parallel as two different answers to the same next question: stateful inspection still can't see payload content, so how do you get real visibility into what's actually being sent? One chose lighter session-layer validation, the other chose heavier, full protocol-aware inspection.
- NGFW emerged once the earlier generations' limitations became genuine business problems at scale. Port-based rules stopped meaningfully reflecting real traffic once almost everything moved to port 443, so organisations needed application awareness, integrated prevention, and visibility into encrypted traffic all at once. NGFW's real innovation is tight integration of these, not any single new idea.
- WAF and UTM are two different responses to the same underlying pressure, more complexity and more specialised threats, but pulling in opposite directions: WAF goes deeper into one specific, high-value area, while UTM broadens outward, bundling many functions for organisations that need adequate protection across many fronts without managing many separate products.
- The overall arc, in one sentence: from matching a header, to tracking a connection, to understanding a protocol, to understanding an application and a user all at once, even inside encryption. Each step trades more processing cost for more context, and therefore stronger, more precise decisions.

SUMMARY

Full Summary Comparison

Type	OSI Layer	Typical Use Case
Packet Filtering	L3/L4, stateless	Basic perimeter or router ACL filtering
Stateful Inspection	L3/L4, stateful	Standard enterprise perimeter firewall
Circuit-Level Gateway	L5 (Session)	Session legitimacy checking, internal address hiding
Application-Layer / Proxy	L7	Deep content control, strong anonymisation
NGFW	L3 to L7, integrated	Modern enterprise perimeter, encrypted traffic inspection
WAF	L7, web-specific	Protecting public-facing web applications
UTM	L3 to L7, bundled modules	SMB or branch-office all-in-one security

Summary

- A firewall enforces an allow/deny policy. An IDS only detects and alerts. An IPS detects and actively blocks, combining both roles.
- Whether a firewall remembers anything about the conversation it's part of, stateless versus stateful, is the single distinction that explains most of the differences between the types covered today.
- The seven types trace a clear arc: from inspecting a header, to tracking a connection, to understanding a protocol, to understanding an application and a user together, even inside encrypted traffic.
- WAF and UTM aren't simply newer than NGFW. They're specialised in opposite directions: WAF goes narrow and deep into web applications specifically, UTM goes broad across many bundled functions.
- Software and hardware firewalls aren't competing choices. Real networks layer both together, exactly like every other defence covered this module.

This Week's Lab & Tutorial

- Deck B, coming next, covers where firewalls actually get placed in a network (including the DMZ and Zone-Based Firewall configuration model), followed by hands-on ACL configuration: Standard, Extended, and Named.
- Tutorial worksheet: study-notes-style questions covering firewall and IDS/IPS definitions, the seven types, and their evolution. Use it to prep for the mid-semester test.

References

- Santos & Stuppi, CCNA Security 210-260 Official Cert Guide, 1st Ed., Cisco Press.
- Kaufman, Perlman, Speciner & Perlner, Network Security: Private Communication in a Public World, 3rd Ed., Prentice Hall.
- Cisco Networking Academy, CCNA Security curriculum (Firewall technologies chapters).